Vectors and Vector-Valued Functions

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Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: 3D Vectors, Dot and Cross Products

Consider the region R in the first quadrant bounded by the curves $y = \sqrt{x}$, $y = 0$, and $x = 4$. A solid is formed by revolving R about the line $x = 5$. Determine the volume of this solid using the method of cylindrical shells. The volume can be expressed in the form $V = k\pi\int_a^b (5 - x)\sqrt{x}\,dx$. Find the value of k.

2 [Foundational] • Target: Derivatives and Integrals of Vector-Valued Functions

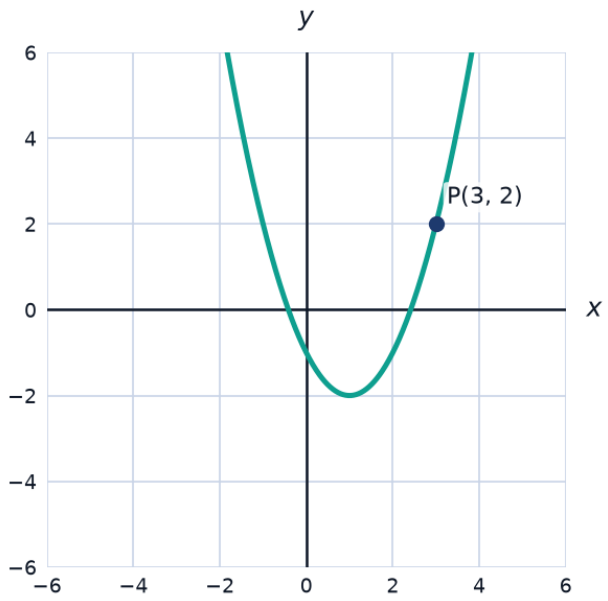
Consider the vector $\mathbf{v}$ with initial point $P = (1, 2)$ and terminal point $Q = (4, 6)$. What are the components of $\mathbf{v}$?

3 [Foundational] • Target: Arc Length and Curvature of Space Curves

Find the curvature of $\mathbf{r}(t) = \langle 3\cos(t), 3\sin(t), 3t\rangle$ at $t = \pi$.

4 [Foundational] • Target: Velocity, Acceleration, and Projectile Motion

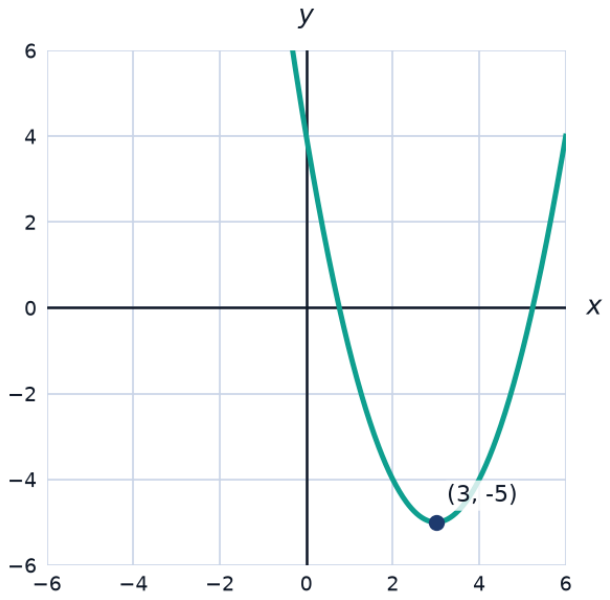
The figure shows the graph of the function $f(x) = x^2 - 2x - 1$ and the point $P(3, 2)$ on the curve. What is the slope of the tangent line to the graph of f at point P?



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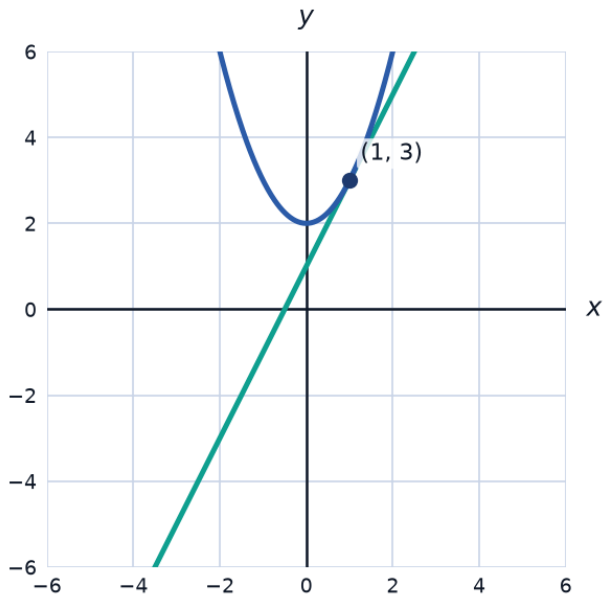
5 [Foundational] • Target: 3D Vectors, Dot and Cross Products

The graph of the quadratic function $f(x) = x^2 - 6x + k$ is shown. The vertex of the parabola is at the point $(3, -5)$. Find the value of the parameter k.



6 [Foundational] • Target: Derivatives and Integrals of Vector-Valued Functions

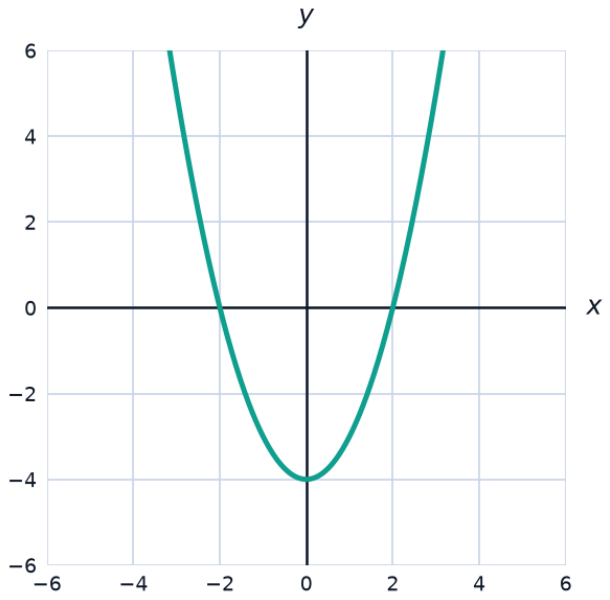
The graph of a differentiable function $f(x)$ is shown below. The function has a tangent line at $x = 1$ with equation $y = 2x + c$. If $f(1) = 3$, find the value of c.



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7 [Foundational] • Target: Arc Length and Curvature of Space Curves

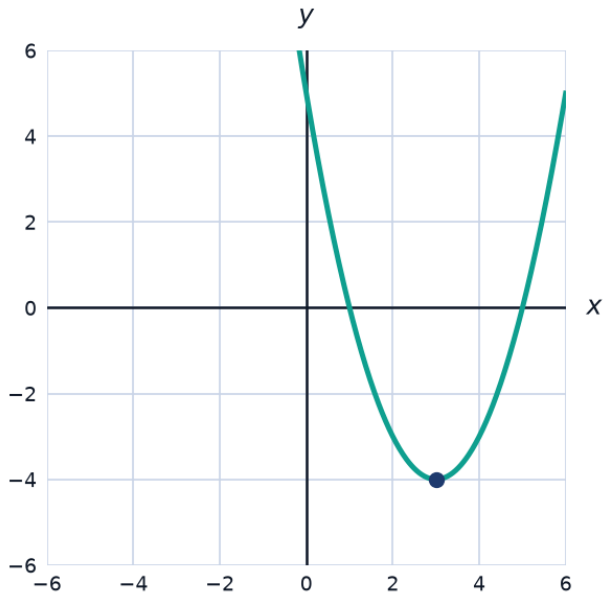
The region bounded by the curve $y = x^2 - 4$ and the x-axis is rotated about the x-axis. If the volume of the resulting solid is $V = \pi\int_a^b (x^2 - 4)^2 dx$, find the value of $b - a$.



Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Velocity, Acceleration, and Projectile Motion

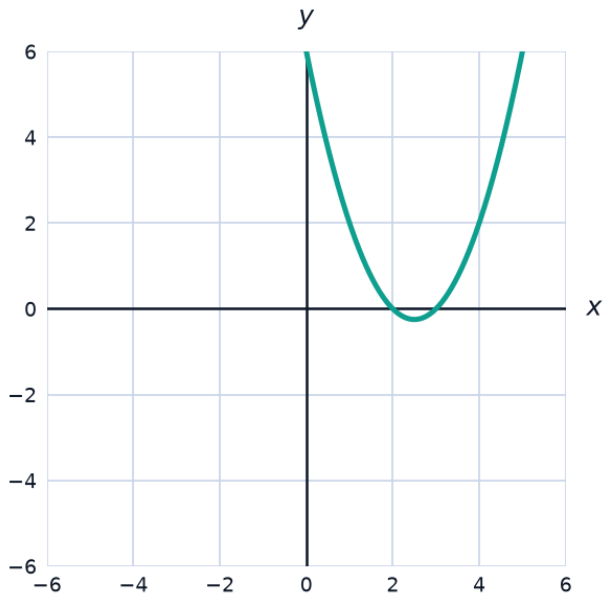
Consider the function $f(x) = x^2 - 6x + 5$. The function has a local minimum at which value of x?



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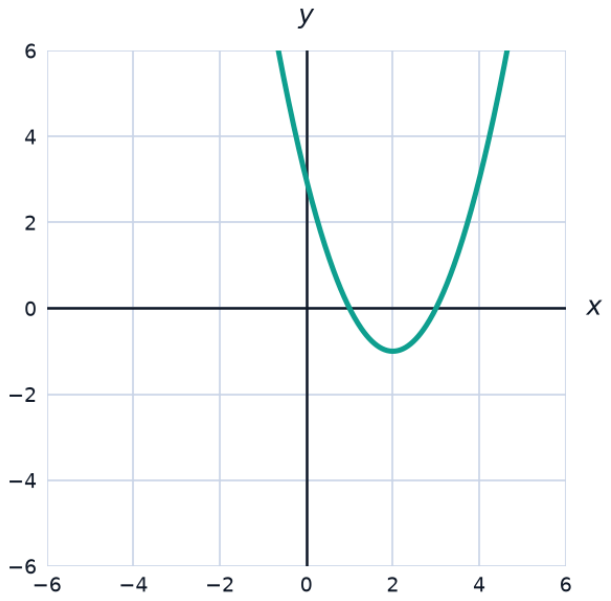
9 **[Intermediate]** • **Target: 3D Vectors, Dot and Cross Products**

Consider the function $f(x) = x^2 - 5x + 6$. What is the derivative of $f(x)$ with respect to x?



10 **[Intermediate]** • **Target: Derivatives and Integrals of Vector-Valued Functions**

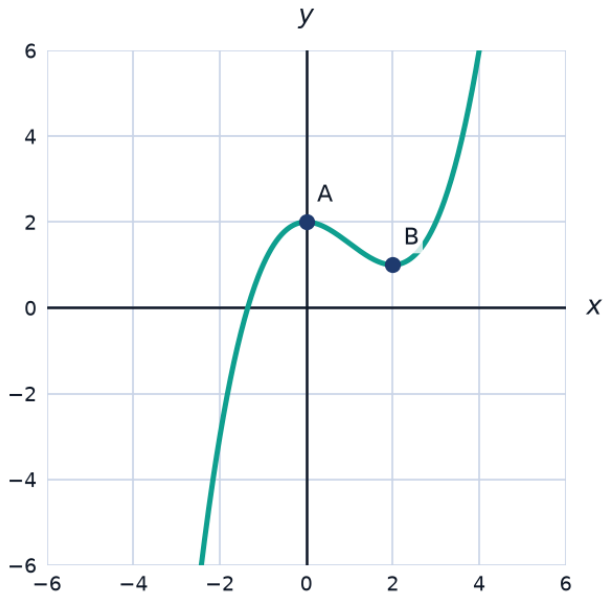
The graph of the function $f(x) = x^2 - 4x + 3$ is shown below. The function has a local minimum. What are the coordinates of the local minimum?



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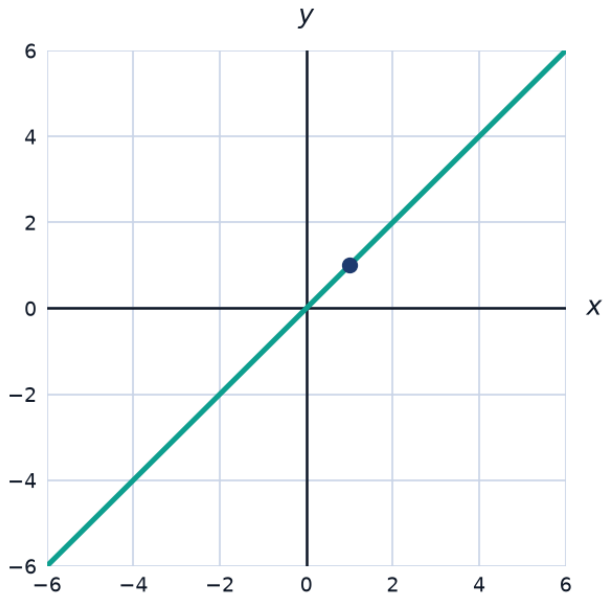
11 **[Intermediate]** • **Target: Arc Length and Curvature of Space Curves**

Consider the arithmetic sequence with the first term $a_1 = 5$ and a common difference $d = 3$. What is the value of the 5th term, a_5?



12 **[Intermediate]** • **Target: Velocity, Acceleration, and Projectile Motion**

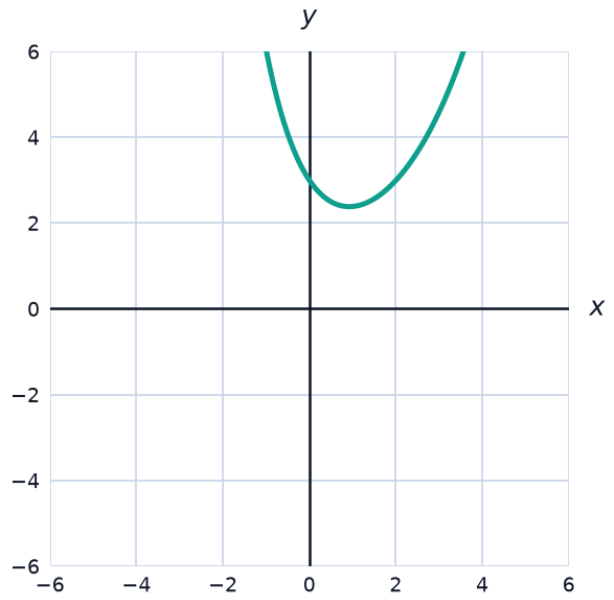
Consider the graph of the function $f(x)$ shown below. What is the limit of $f(x)$ as x approaches 1?



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13 [Intermediate] • Target: 3D Vectors, Dot and Cross Products

Solve $y' + 2y = x$, $y(0) = -3$. Find $y(1)$.



14 [Intermediate] • Target: Derivatives and Integrals of Vector-Valued Functions

Consider the function $z = f(x, y) = x^2 + 2y^2$. What is the value of the partial derivative $\frac{\partial f}{\partial x}$ at the point $(1, 2)$?

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Arc Length and Curvature of Space Curves • Cross-Lesson: 8.01 Area Between Curves

Consider the region bounded by the curve $y = x^2$, the x-axis, and the vertical lines $x = 1$ and $x = 3$. What is the area of this region?

16 [Challenging] • Target: Velocity, Acceleration, and Projectile Motion • Cross-Lesson: 8.02 Volumes of Revolution

$\frac{d}{dx}\left[\int_{\cos x}^{\sin x} e^{t^2}\, dt\right] =$

17 [Challenging] • Target: 3D Vectors, Dot and Cross Products • Cross-Lesson: 8.03 Arc Length and Improper Integrals

Consider the function $z = f(x, y) = x^2 - 3xy + 2y^2$. What is the partial derivative of f with respect to y, denoted as $\frac{\partial f}{\partial y}$?

18 [Challenging] • Target: Derivatives and Integrals of Vector-Valued Functions • Cross-Lesson: 9.01 Separable Differential Equations

A particle's position in the xy-plane is given by the parametric equations $x(t) = 2t + 1$ and $y(t) = t^2 - 3$, where t is time. What is the Cartesian equation of the path traced by the particle?

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19 [Challenging] • Target: Arc Length and Curvature of Space Curves • Cross-Lesson: 1.01 Precalculus Review for Calculus

A particle's position is described by the parametric equations $x(t) = 3t + 1$ and $y(t) = 2t - 2$. What are the coordinates of the particle when $t = 3$?

20 [Challenging] • Target: Velocity, Acceleration, and Projectile Motion • Cross-Lesson: 10.01 Calculus with Parametric and Polar Curves

Reduction formula: $\int \sin^n x\,dx = -\frac{\sin^{n-1} x \cos x}{n} + \frac{n-1}{n}\int \sin^{n-2} x\,dx$. Then $\int_0^{\pi/2} \sin^6 x\,dx =$

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: 3D Vectors, Dot and Cross Products

Given $\mathbf{u} = \langle 2, -1, 3 \rangle$ and $\mathbf{v} = \langle 4, 2, -1 \rangle$, find $\mathbf{u} \cdot \mathbf{v}$.

a. 3

b. 7

c. 13

d. -1

22 [Foundational] • Target: 3D Vectors, Dot and Cross Products

Find the cross product $\mathbf{u} \times \mathbf{v}$ for $\mathbf{u} = \langle 1, 0, 2 \rangle$ and $\mathbf{v} = \langle 0, 3, -1 \rangle$.

a. $\langle -6, 1, 3 \rangle$

b. $\langle 6, -1, -3 \rangle$

c. $\langle -6, -1, 3 \rangle$

d. $\langle 6, 1, 3 \rangle$

23 [Foundational] • Target: 3D Vectors, Dot and Cross Products

Find the area of the parallelogram determined by $\mathbf{u} = \langle 1, 2, 0 \rangle$ and $\mathbf{v} = \langle 0, 3, 4 \rangle$.

a. $\sqrt{89}$

b. 9

c. $\sqrt{73}$

d. 10

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24 [Foundational] • Target: Derivatives and Integrals of Vector-Valued Functions

Given $\mathbf{r}(t) = \langle t^2, \sin(2t), e^{3t} \rangle$, find $\mathbf{r}'(t)$.

a. $\langle 2t, 2\cos(2t), 3e^{3t} \rangle$

b. $\langle 2t, \cos(2t), e^{3t} \rangle$

c. $\langle t, 2\cos(2t), 3e^{3t} \rangle$

d. $\langle 2t, -2\cos(2t), 3e^{3t} \rangle$

25 [Foundational] • Target: Derivatives and Integrals of Vector-Valued Functions

Find $\int_0^1 \langle 2t, 3t^2, 4t^3 \rangle \, dt$.

a. $\langle 1, 1, 1 \rangle$

b. $\langle 2, 3, 4 \rangle$

c. $\langle 1/2, 1/3, 1/4 \rangle$

d. $\langle 1, 2, 3 \rangle$

26 [Foundational] • Target: Derivatives and Integrals of Vector-Valued Functions

Find the unit tangent vector $\mathbf{T}(t)$ to $\mathbf{r}(t) = \langle 3\cos t, 3\sin t, 4t \rangle$.

a. $\langle -\frac{3}{5}\sin t, \frac{3}{5}\cos t, \frac{4}{5} \rangle$

b. $\langle -3\sin t, 3\cos t, 4 \rangle$

c. $\langle -\frac{3}{4}\sin t, \frac{3}{4}\cos t, 1 \rangle$

d. $\frac{1}{5}\langle 3\cos t, 3\sin t, 4t \rangle$

27 [Foundational] • Target: Arc Length and Curvature of Space Curves

Find the arc length of the circular helix $\mathbf{r}(t) = \langle 3\cos t, 3\sin t, 4t \rangle$ for $0 \le t \le 2\pi$.

a. 10π

b. 5π

c. 20π

d. 6π

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28 [Foundational] • Target: Arc Length and Curvature of Space Curves

State the formula for curvature $\kappa(t)$ of a space curve in terms of $\mathbf{r}'$ and $\mathbf{r}''$.

a. $\kappa(t) = \frac{||\mathbf{r}'(t) \times \mathbf{r}''(t)||}{||\mathbf{r}'(t)||^3}$

b. $\kappa(t) = \frac{||\mathbf{r}''(t)||}{||\mathbf{r}'(t)||^2}$

c. $\kappa(t) = \frac{\mathbf{r}'(t) \cdot \mathbf{r}''(t)}{||\mathbf{r}'(t)||^3}$

d. $\kappa(t) = ||\mathbf{r}''(t)||$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Arc Length and Curvature of Space Curves

Find the curvature of a circle of radius R.

a. $\kappa = 1/R$

b. $\kappa = R$

c. $\kappa = 2\pi R$

d. $\kappa = 1/R^2$

30 [Intermediate] • Target: Velocity, Acceleration, and Projectile Motion

If position is $\mathbf{r}(t) = \langle t^2, 4t, -16t^2 + 64t \rangle$, find the velocity and acceleration vectors at $t = 1$.

a. $\mathbf{v}(1) = \langle 2, 4, 32 \rangle, \mathbf{a}(1) = \langle 2, 0, -32 \rangle$

b. $\mathbf{v}(1) = \langle 1, 4, 48 \rangle, \mathbf{a}(1) = \langle 2, 0, -16 \rangle$

c. $\mathbf{v}(1) = \langle 2, 4, 0 \rangle, \mathbf{a}(1) = \langle 2, 0, -32 \rangle$

d. $\mathbf{v}(1) = \langle 2, 4, -32 \rangle$

31 [Intermediate] • Target: Velocity, Acceleration, and Projectile Motion

Find the speed at $t = 0$ for a particle with velocity $\mathbf{v}(t) = \langle 3\cos t, 4, 3\sin t \rangle$.

a. 5

b. 7

c. 4

d. $\sqrt{7}$

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32 [Intermediate] • Target: Velocity, Acceleration, and Projectile Motion

Decompose acceleration into tangential and normal components: $\mathbf{a} = a_T\mathbf{T} + a_N\mathbf{N}$. What is a_T?

a. $a_T = \frac{\mathbf{v} \cdot \mathbf{a}}{||\mathbf{v}||}$

b. $a_T = \frac{||\mathbf{v} \times \mathbf{a}||}{||\mathbf{v}||}$

c. $a_T = ||\mathbf{a}||$

d. $a_T = \frac{\mathbf{a}}{||\mathbf{v}||}$

33 [Intermediate] • Target: 3D Vectors, Dot and Cross Products

Find the arc length of $\mathbf{r}(t) = \langle 2t, t^2, \ln(t) \rangle$ for $1 \le t \le 3$.

a. $2\ln(3) + 16$

b. $\ln(3) + 8$

c. $\frac{\ln(3)}{2} + 4$

d. $\ln(3) + 9$

34 [Intermediate] • Target: Derivatives and Integrals of Vector-Valued Functions

Find the arc length of $\mathbf{r}(t) = \langle 2t, t^2, \ln(t) \rangle$ for $1 \le t \le 2$.

a. $2\ln(2) + 6$

b. $\ln(2) + 3$

c. $\frac{\ln(2)}{2} + \frac{3}{2}$

d. $\ln(2) + 4$

35 [Intermediate] • Target: Arc Length and Curvature of Space Curves

Find the arc length of $\mathbf{r}(t) = \langle e^t, e^{-t}, \sqrt{2}\,t \rangle$ for $0 \le t \le \ln(2)$.

a. $\frac{3}{2}$

b. 3

c. $\frac{3}{4}$

d. $\frac{5}{2}$

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36 [Intermediate] • Target: Velocity, Acceleration, and Projectile Motion

Find the arc length of $\mathbf{r}(t) = \langle e^t, e^{-t}, \sqrt{2}\,t \rangle$ for $0 \le t \le \ln(3)$.

a. $\frac{16}{3}$

b. $\frac{8}{3}$

c. $\frac{4}{3}$

d. $\frac{11}{3}$

37 [Intermediate] • Target: 3D Vectors, Dot and Cross Products

Arc length of $\mathbf{r}(t) = (e^t \cos t, e^t \sin t, e^t)$ from $t = 0$ to $t = 1$:

a. $e - 1$

b. $\sqrt{3}(e - 1)$

c. $\sqrt{2}(e - 1)$

d. $e\sqrt{3}$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Derivatives and Integrals of Vector-Valued Functions • Cross-Lesson: 9.01 Separable Differential Equations

Arc length of $\mathbf{r}(t) = (t, t, t)$ from $t = 0$ to $t = 1$:

a. 1

b. $\sqrt{2}$

c. $\sqrt{3}$

d. 3

39 [Challenging] • Target: Arc Length and Curvature of Space Curves • Cross-Lesson: 1.01 Precalculus Review for Calculus

Find the curvature of $\mathbf{r}(t) = \langle e^t, e^{-t}, \sqrt{2}\,t \rangle$ at $t = 0$.

a. $\frac{\sqrt{2}}{2}$

b. $\sqrt{2}$

c. $\frac{\sqrt{2}}{4}$

d. $2\sqrt{2}$

40 [Challenging] • Target: Velocity, Acceleration, and Projectile Motion • Cross-Lesson: 10.01 Calculus with Parametric and Polar Curves

Find the curvature of $\mathbf{r}(t) = \langle 3\cos(t), 3\sin(t), 3t \rangle$ at $t = \frac{\pi}{2}$.

a. $\frac{1}{6}$

b. $\frac{\sqrt{2}}{2}$

c. 3

d. $\frac{1}{3}$

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41 [Challenging] • Target: 3D Vectors, Dot and Cross Products • Cross-Lesson: 11.01 Sequences and Series

Find the curvature of $\mathbf{r}(t) = \langle 3\cos(t), 3\sin(t), 4t \rangle$ at $t = \pi$.

a. $\frac{3}{25}$

b. $\frac{3}{5}$

c. 3

d. $\frac{6}{25}$

42 [Challenging] • Target: Derivatives and Integrals of Vector-Valued Functions • Cross-Lesson: 12.01 Convergence Tests

Find the curvature of $\mathbf{r}(t) = \langle \cos(t), \sin(t), 2t \rangle$ at $t = 0$.

a. $\frac{\sqrt{5}}{5}$

b. 1

c. $\frac{2}{5}$

d. $\frac{1}{5}$

43 [Challenging] • Target: Arc Length and Curvature of Space Curves • Cross-Lesson: 12.02 Power Series and Taylor Series

Find the curvature of $\mathbf{r}(t) = (t, t^2, 0)$ at $t = 0$:

a. 2

b. 1

c. 0

d. $1/2$

44 [Challenging] • Target: Velocity, Acceleration, and Projectile Motion • Cross-Lesson: 14.01 Partial Derivatives and the Gradient

For $\mathbf{r}(t) = \langle 2t, 3t^2, t \rangle$, find the normal component of acceleration at $t = 1$.

a. $\frac{6\sqrt{205}}{41}$

b. $\frac{36\sqrt{41}}{41}$

c. 6

d. 36

45 [Challenging] • Target: 3D Vectors, Dot and Cross Products • Cross-Lesson: 15.01 Double Integrals

For $\mathbf{r}(t) = \langle 2t, 2t^2, t \rangle$, find the normal component of acceleration at $t = 1$.

a. $\frac{16\sqrt{21}}{21}$

b. 4

c. 16

d. $\frac{4\sqrt{105}}{21}$

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